MULTIMODAL EMBEDDINGS

From CLIP to Gemini Embedding 2

Architecture, mathematics, training and retrieval evaluation

EXTENDED RESEARCH EDITION 2020–2026

15 model families • Vector architecture schematics • Properly typeset LaTeX • Fully solved examples • Primary-paper links

Core intuition

Learning method: For every model ask: (1) what was missing? (2) where do the tokens flow? (3) what loss trains it? (4) can you calculate a miniature example? (5) what still fails? All toy scores in this guide are explicitly illustrative, not measurements generated by an actual trained model.

Prepared as an independent educational reconstruction, October 23, 2026. All original papers are linked. Drawings are not official architecture figures. Model weights and paper text can carry different licenses.

Revision: v2.0 | October 23, 2026 | Companion to prior 14-page handbook.

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1. Start with the map: what changed between 2020 and 2026?

Year	Milestone	Architectural / training shift
2020	SimCLR	Image-only two-view self-supervised contrastive training.
2021	CLIP	Two modality-specific towers into one image–text retrieval space.
2021	ALIGN	Scale noisy paired supervision dramatically; dual-encoder recipe.
2021	AudioCLIP	Add a separate audio encoder into CLIP's image/text space.
2021	VideoCLIP	Align temporal video segments and text with hard negative mining.
2022	CLAP	Direct audio–language alignment for zero-shot sound retrieval.
2022	CoCa	Contrastive retrieval features and autoregressive caption generation.
2023	SigLIP	Pairwise sigmoid replaces global contrastive softmax.
2023	ImageBind	Six modalities aligned via image-paired training; not all pairs required.
2024	ColPali	Many page-patch vectors; MaxSim late interaction for visual documents.
2024	MM-Embed	Adapt multimodal LLM into retriever; modality-aware hard negatives.
2025	SigLIP 2	Dense localization, multilingual vision–language learning, multiple objectives.
2025	Gemini Embedding	Retrieval-specialized encoder adapted from pretrained Gemini; text only.
2026	Gemini Embedding 2	Interleaved multimodal token fusion, bidirectional transformer, pooling, MRL.
2026	EmbeddingGemma 2	Small, downloadable multimodal embedder with local deployment.

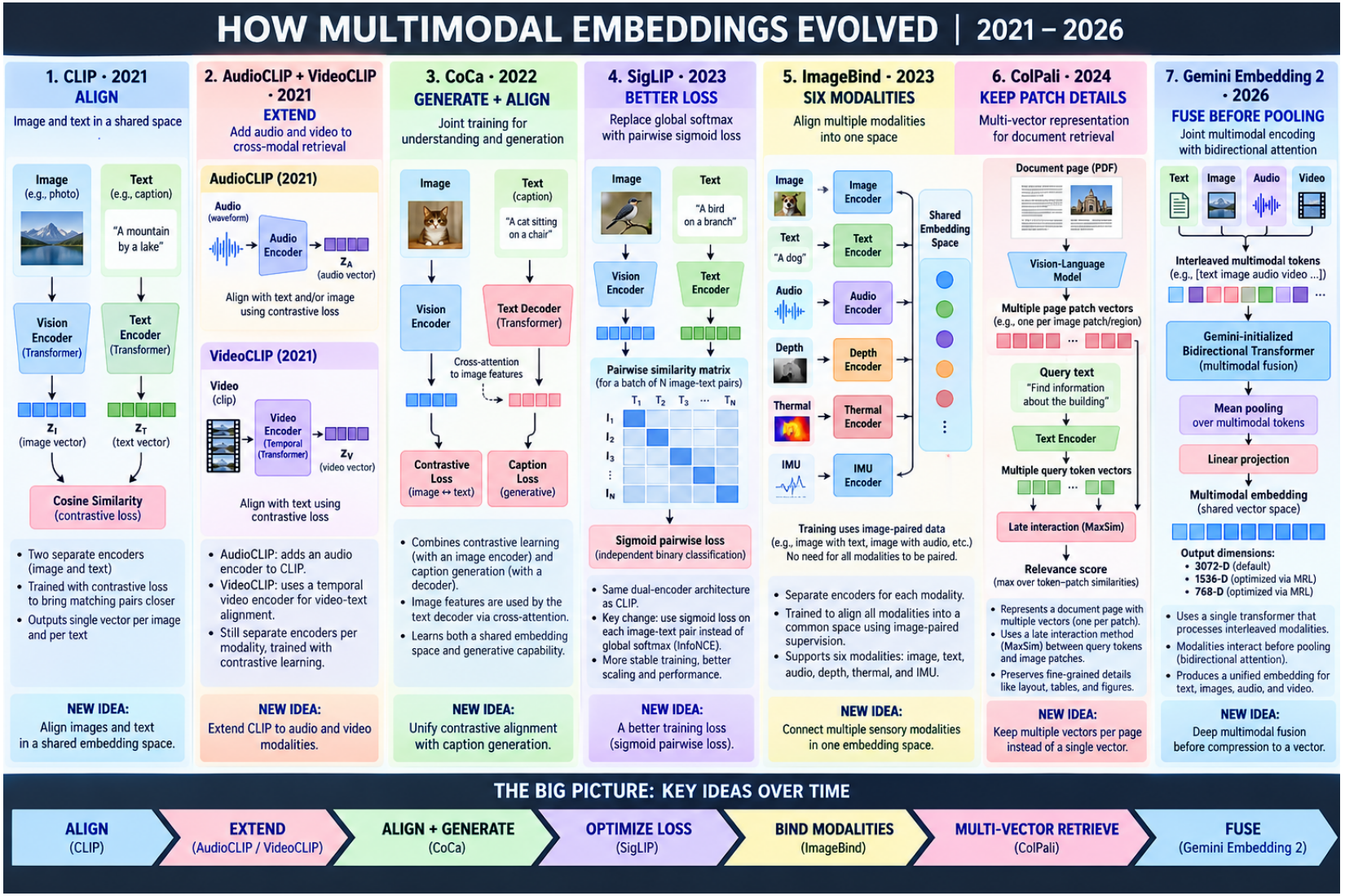
Research caveat

History is not a straight replacement chain: CoCa, SigLIP, ImageBind, ColPali and Gemini address *different* bottlenecks. Also, VideoCLIP and AudioCLIP were not the first-ever audio/video representation models. The dates above follow original papers; MM-Embed first appeared in 2024 (revised 2025).

1.1 Your previously generated memory map

The illustration on the following page was generated in our earlier conversation. It is a conceptual study aid; the vector diagrams in the subsequent sections are newly drawn, scalable diagrams and are more specific about the mechanisms.

Architecture memory map (previously generated)



Instructional illustration; not a figure extracted from the cited papers.

2. The mathematics you will keep reusing

2.1 An embedding is a learned representation, not a fact database

An encoder maps an input x to $z = f_\theta(x) \in \mathbb{R}^d$. The training objective determines what "near" means: two audio clips may be close by sound, or a caption and image may be close by semantics. Vector coordinates are not universally meaningful across independently trained models.

$$\hat{z} = \frac{z}{\|z\|_2}, \qquad \cos(q,p) = \frac{q^\top p}{\|q\|_2\|p\|_2}. \tag{1}$$

Worked example (illustrative, not model measurements)

Let $q = [3,4]$, $p = [0,5]$. Compute $q^\top p = 3(0) + 4(5) = 20$; $\|q\|_2 = \sqrt{3^2+4^2} = 5$; $\|p\|_2 = 5$. Hence

$$\cos(q,p) = \frac{20}{5\times 5} = \boxed{0.8}.$$

This is a similarity score, *not* 80% Recall@1, and not a guaranteed 80% probability.

2.2 Contrastive alignment: turn matching pairs into geometry

In a batch of B matched pairs, the normalized image–text similarity matrix is

$$S_{ij} = \frac{\hat{v}_i^\top \hat{t}_j}{\tau}, \qquad \mathscr{L}_{i\to t} = -\frac{1}{B}\sum_{i=1}^{B} \log \frac{e^{S_{ii}}}{\sum_{j=1}^{B} e^{S_{ij}}}. \tag{2}$$

CLIP uses both directions: $\mathscr{L}_{\text{CLIP}} = \frac{1}{2}(\mathscr{L}_{i\to t} + \mathscr{L}_{t\to i})$. The temperature τ changes how sharply the softmax distinguishes candidates.

Worked example (illustrative, not model measurements)

For a query with one positive at cosine 0.8 and two negatives at $0.5, 0.2$, choose $\tau = 0.1$:

$$P_+ = \frac{e^8}{e^8 + e^5 + e^2} \approx 0.9503, \quad \mathscr{L} = -\log P_+ \approx \boxed{0.0511}.$$

Steps: (1) divide each similarity by 0.1, obtaining logits $8, 5, 2$; (2) exponentiate; (3) normalize; (4) take the negative natural log.

2.3 Pairwise sigmoid: learn without row-wise softmax

A simplified sign-based sigmoid objective for two independent pairs is

$$\ell(s,y) = \log\left(1 + e^{-ys}\right), \qquad y \in \{+1,-1\}.$$

SigLIP averages/scales such terms over its batch with a learnable logit scale and bias; the displayed equation isolates the fundamental per-pair mechanism.

Worked example (illustrative, not model measurements)

A positive pair receives logit $s = 2$, $y = +1$: $\ell = \log(1 + e^{-2}) = 0.1269$. A negative pair receives logit $s = 0$, $y = -1$: $\ell = \log(2) = 0.6931$. Mean loss $= (0.1269 + 0.6931)/2 = \boxed{0.4100}$. Each pair is judged independently, rather than normalized across a softmax row.

2.4 Attention, pooling, projection

$$Q = HW_Q, \qquad K = HW_K, \qquad V = HW_V, \tag{3}$$

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + M\right)V, \tag{4}$$

$$H' = \mathscr{M}(T) \in \mathbb{R}^{L \times h}, \qquad \bar{h} = \frac{1}{L}\sum_{i=1}^{L} H'_i, \qquad e = W\bar{h} + b. \tag{5}$$

M is an attention mask; a bidirectional representation model does not use the strictly causal upper-triangular mask of a standard autoregressive decoder. Mean pooling reduces a variable-length set of contextualized token states to a fixed-length vector.

Worked example (illustrative, not model measurements)

Four token states are $H' = [(1,0),(0,1),(1,1),(2,0)]$. Their mean is

$$\bar{h} = \frac{(1,0)+(0,1)+(1,1)+(2,0)}{4} = (1, 0.5).$$

Choose a toy projection $W = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$ and $b = (0,0)$. Then $e = W\bar{h} = (2, 0.5)$. Google's real projection is learned, much larger, and these toy values are invented solely for calculation.

2.5 Late interaction and Matryoshka embeddings

$$\text{MaxSim}(q, d) = \sum_{i=1}^{|q|} \max_{1 \leqslant j \leqslant |d|} \widehat{q}_i^\top \widehat{d}_j, \tag{6}$$

$$\mathscr{L}_{\text{MRL}} = \sum_{m \in \mathscr{D}} \lambda_m \, \mathscr{L}_m(z_{1:m}) \,. \tag{7}$$

Worked example (illustrative, not model measurements)

Two query tokens score two page patches as $[0.8, 0.3]$ and $[0.1, 0.9]$. MaxSim chooses each row's best patch: $0.8 + 0.9 = \boxed{1.7}$. Unlike mean pooling, this preserves which region matches each query token. For MRL, a 768D prefix of a 3072D vector is separately supervised, not merely truncated arbitrarily after ordinary training.

3. Architectures and training: model-by-model laboratory

Each schematic is an *educational abstraction* of the published mechanism; it does not imply identical modality encoders, vector sizes, or paper-specific operations beyond what the source reports.

3.1 SimCLR (2020)

Original source: A Simple Framework for Contrastive Learning of Visual Representations ↗

Problem: Learn image features without category labels. **Mechanism:** two random augmentations of one image, shared encoder f, and projection head g; a temperature-scaled NT-Xent contrastive objective chooses the correct second view among the other views. At downstream inference the projection head may be discarded.



$$\ell_{i,j} = -\log \frac{\exp(\mathrm{sim}(z_i, z_j)/\tau)}{\sum_{k \neq i} \exp(\mathrm{sim}(z_i, z_k)/\tau)}. \tag{8}$$

Worked example (illustrative, not model measurements)

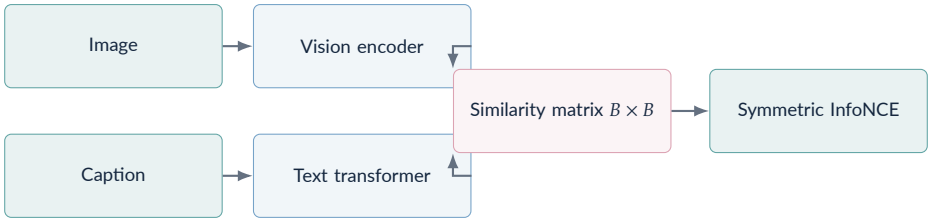
1. Crop and recolor a cat photograph twice: views a and b. A third picture is a truck c.
2. Suppose $\mathrm{sim}(a,b) = 0.9$, $\mathrm{sim}(a,c) = 0.1$, and $\tau = 0.2$.
3. Softmax on candidates b, c: $P(b \mid a) = e^{4.5}/(e^{4.5} + e^{0.5}) \approx 0.9820$.
4. Loss $= -\ln(0.9820) \approx \boxed{0.0181}$. This learns invariance to the crop and recoloring.

Recall: SAME IMAGE + TWO VIEWS → REPRESENTATION WITHOUT LABELS. Limitation: only image representations; augmentation choices can accidentally erase details a later task needs.

3.2 CLIP (2021)

Original source: Learning Transferable Visual Models From Natural Language Supervision ↗

Problem: Image classifiers tied to a fixed label set. **Architecture:** a visual encoder and text transformer with learned projections to one space; symmetric image-to-text and text-to-image contrastive loss, trained on hundreds of millions of image–text pairs in the original work.



$$\mathscr{L}_{\mathrm{CLIP}} = \frac{1}{2}\left[-\frac{1}{B}\sum_i \log \frac{e^{S_{ii}}}{\sum_j e^{S_{ij}}} - \frac{1}{B}\sum_i \log \frac{e^{S_{ii}}}{\sum_j e^{S_{ji}}} \right]. \tag{9}$$

Worked example (illustrative, not model measurements)

1. Use two photos (cat, truck) and two descriptions ("a cat", "a truck"). After normalization, suppose the similarity matrix is
$$C = \begin{bmatrix} 0.8 & 0.2 \\ 0.1 & 0.9 \end{bmatrix}.$$
2. With $\tau = 0.2$, row 1 logits become $[4, 1]$, so $P(\text{cat caption} \mid \text{cat image}) = e^4/(e^4 + e^1) = 0.9526$.
3. Row 2 logits are $[0.5, 4.5]$, so $P(\text{truck caption} \mid \text{truck image}) = 0.9820$.
4. For the reverse text-to-image direction, compute column softmaxes as well. Average the four neg-

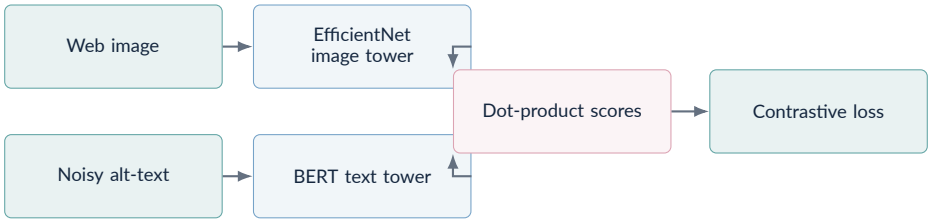
ative log-likelihoods: $\mathscr{L} \approx \boxed{0.0316}$ (rounded).

Recall: TWO TOWERS → MATCH IMAGE TO TEXT. CLIP does not jointly encode arbitrary text+image instructions before creating the final embedding. A two-tower approach is fast because corpus vectors can be indexed once.

3.3 ALIGN (2021)

Original source: Scaling Up Visual and Vision-Language Representation Learning With Noisy Text Supervision ↗

Problem: Curating perfectly clean paired images and captions at huge scale is expensive. **Mechanism:** keep the dual-encoder contrastive approach but scale to a reported 1.8 billion noisy image–alt-text pairs, demonstrating strong zero-shot/retrieval results without exhaustive filtering.



$$S_{ij} = \frac{v_i^\top t_j}{\tau}, \quad \mathscr{L} \text{ is a bidirectional softmax contrastive objective.} \tag{10}$$

Worked example (illustrative, not model measurements)

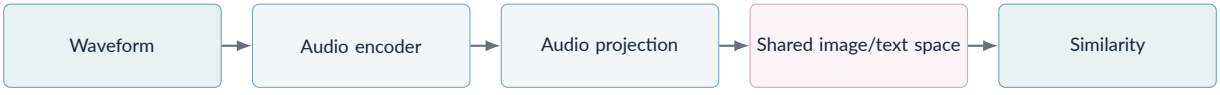
1. Suppose a noisy caption says “brown animal grass” for a deer photograph, while a clean description says “deer walking on grass”.
2. An illustrative positive logit is 3.0 and the unrelated negative logit is 1.0. Then $P_+ = e^3/(e^3 + e^1) = 0.8808$.
3. Loss = $-\ln(0.8808) = \boxed{0.1269}$; many varied pairs teach shared concepts despite individual noisy captions.

Recall: ALIGN → SCALE THE PAIRS, NOT A NEW FUSION ARCHITECTURE. More web data also brings bias, duplicates, and mismatched supervision.

3.4 AudioCLIP (2021)

Original source: AudioCLIP: Extending CLIP to Image, Text and Audio ↗

Problem: CLIP had no native sound encoder. **Mechanism:** connect an ESResNeXt-based audio branch to the CLIP image and text encoders; train alignment so sound, caption, and image can support cross-modal retrieval and classification.



$$a = f_a(\text{waveform}), \quad t = f_t(\text{caption}), \quad S = \cos(a, t). \tag{11}$$

Worked example (illustrative, not model measurements)

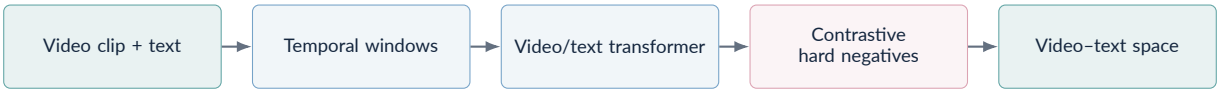
1. Query audio: a dog barking. Candidate captions: “dog barking”, “car engine”.
2. Suppose normalized similarities are $[0.75, 0.15]$ and $\tau = 0.2$.
3. $P(\text{dog} \mid \text{audio}) = e^{3.75}/(e^{3.75} + e^{0.75}) = 0.9526$.
4. Training loss = $-\ln(0.9526) = \boxed{0.0486}$.

Recall: AUDIOCLIP → ADD A THIRD ENCODER TO THE ALIGNED SPACE. Distinct from learning joint audio+image fusion in a single transformer.

3.5 VideoCLIP (2021)

Original source: VideoCLIP: Contrastive Pre-training for Zero-shot Video-Text Understanding ↗

Problem: Static-image similarity is insufficient for events unfolding through time. **Mechanism:** contrast temporally overlapping video and text clips; hard negatives include semantically close retrieved examples. Published architecture is temporal video–text representation learning, not simply feeding a video file into a static CLIP encoder.



$$\ell = -\log \frac{\exp(s(v^+, t)/\tau)}{\exp(s(v^+, t)/\tau) + \sum_{v^-} \exp(s(v^-, t)/\tau)}. \tag{12}$$

Worked example (illustrative, not model measurements)

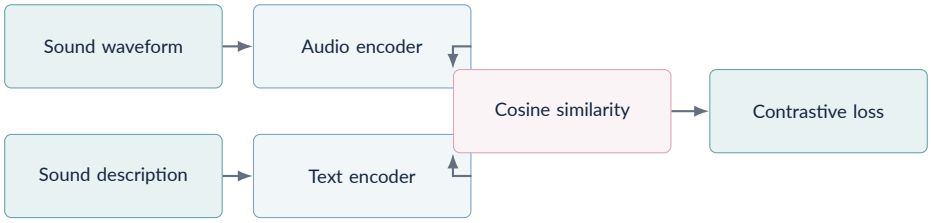
1. Query text "person opens a door". Positive video window shows the door opening. Hard negative shows someone standing near the same door.
2. Set positive similarity 0.7, hard-negative 0.6, and unrelated video 0.1; $\tau = 0.1$.
3. $P_+ = e^7/(e^7 + e^6 + e^1) \approx 0.7296$; loss $\approx \boxed{0.3153}$.
4. The near-match is hard because appearance overlaps, but the action differs.

Recall: VIDEOCLIP → ALIGN TIME-DEPENDENT EVENTS WITH TEXT. Temporal sampling quality matters.

3.6 CLAP (2022)

Original source: CLAP: Learning Audio Concepts From Natural Language Supervision ↗

Problem: Fixed audio classifiers cannot understand arbitrary user descriptions. **Mechanism:** audio and language encoders are contrastively trained on audio–text pairs. The original study reports 128K pairs, permitting flexible zero-shot audio classification/retrieval.



$$P(t_i \mid a_i) = \frac{\exp(\mathrm{sim}(a_i, t_i)/\tau)}{\sum_j \exp(\mathrm{sim}(a_i, t_j)/\tau)}. \tag{13}$$

Worked example (illustrative, not model measurements)

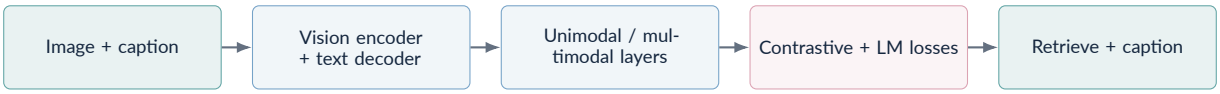
1. Query sound: rain on a metal roof. Captions: "heavy rain", "door closes", "motor starts".
2. Similarities are $0.65, 0.25, 0.05$ at $\tau = 0.2$, yielding logits $3.25, 1.25, 0.25$.
3. $P(\text{rain}) = e^{3.25}/(e^{3.25} + e^{1.25} + e^{0.25}) \approx 0.8438$.
4. Negative log-loss $\approx \boxed{0.1699}$.

Recall: CLAP → AUDIO ↔ LANGUAGE, LIKE CLIP FOR SOUND. It does not automatically understand native video sequences.

3.7 CoCa (2022)

Original source: CoCa: Contrastive Captioners are Image-Text Foundation Models ↗

Problem: Contrastive alignment alone does not teach a model to generate detailed language about an image. **Architecture:** visual encoder + multimodal text decoder with unimodal and cross-attention portions. Jointly optimize a global contrastive representation loss and a caption generation loss.



$$\mathcal{L} = \lambda_c \mathcal{L}_{\text{contrast}} + \lambda_g \mathcal{L}_{\text{caption}}, \quad \mathcal{L}_{\text{caption}} = -\sum_t \log P(y_t \mid y_{<t}, I). \tag{14}$$

Worked example (illustrative, not model measurements)

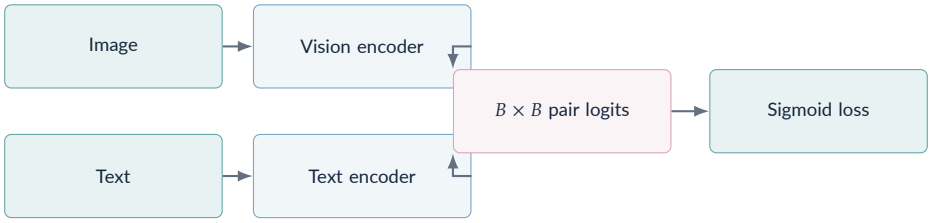
1. Suppose contrastive loss is 0.30 and the teacher-forced caption contains two tokens with predicted probabilities 0.8 and 0.5.
2. Caption loss $= -\ln 0.8 - \ln 0.5 = 0.2231 + 0.6931 = 0.9163$.
3. With a toy equal-weight combination $\lambda_c = \lambda_g = 1$, total loss $0.30 + 0.9163 = \boxed{1.2163}$. Actual training hyperparameters must be taken from the paper.

Recall: CoCa → RETRIEVE **AND** GENERATE. Its decoder adds complexity; this is not the same as a pure retrieval bi-encoder.

3.8 SigLIP (2023)

Original source: Sigmoid Loss for Language Image Pre-Training ↗

Problem: CLIP softmax contrasts matches against all candidates in a global batch. **Contribution:** independently score image–text pairs with a sigmoid objective. The fundamental image/text dual-tower idea is retained; the objective changes.



$$\ell_{ij} = \log\left[1 + \exp\left(-y_{ij}(\alpha v_i^\top t_j + \beta)\right)\right], \quad y_{ij} \in \{-1, +1\}. \tag{15}$$

Worked example (illustrative, not model measurements)

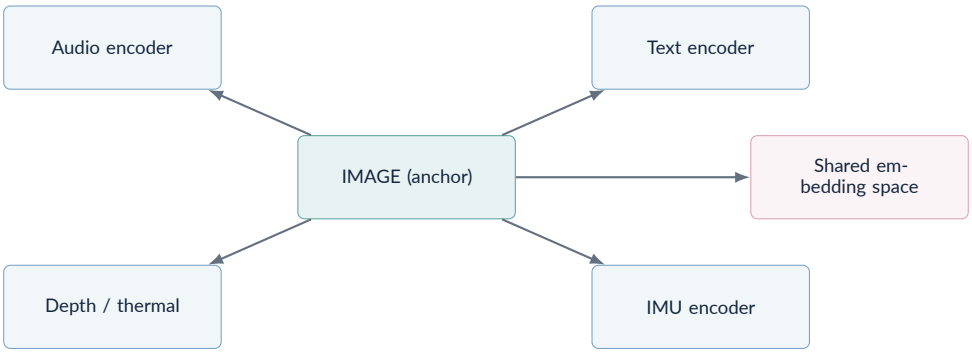
1. Use 2 matched image–text examples: 2 positive pairs and 2 mismatches. For pedagogical simplicity set $\alpha = 1, \beta = 0$.
2. Set positive logits $2, 1$ and negative logits $-1, 0$. Pair losses are $0.1269, 0.3133, 0.3133, 0.6931$.
3. Average $= (0.1269 + 0.3133 + 0.3133 + 0.6931)/4 = \boxed{0.3617}$.
4. Each binary match decision is trained without normalizing its row against all captions.

Recall: SigLIP → THE LOSS CHANGED; NO NEW MODALITY WAS ADDED.

3.9 ImageBind (2023)

Original source: ImageBind: One Embedding Space To Bind Them All ↗

Problem: A shared space among many modalities seems to require audio–text, audio–thermal, text–IMU, and all other expensive pair types. **Idea:** learn separate encoders for image, text, audio, depth, thermal, and IMU; align non-image modalities primarily using image-paired data, so emergent cross-modal retrieval can occur without direct pairing of every modality.



$$\mathscr{L} \approx \sum_{m \neq \text{image}} \lambda_m \, \mathscr{L}_{\text{contrast}}\big(f_{\text{image}}(I), f_m(x_m)\big). \tag{16}$$

Worked example (illustrative, not model measurements)

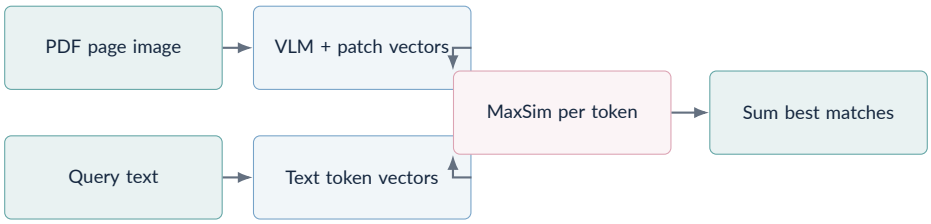
1. Train an audio "dog bark" with its associated dog image and a text caption "a dog" with a matching dog image.
2. Suppose the resulting normalized vectors are $i = (1, 0)$, $a = (0.98, 0.20)/\sqrt{1.0004}$, and $t = (0.96, 0.28)/\sqrt{1.0000}$.
3. Their audio–text cosine is approximately $0.98(0.96) + 0.20(0.28) = \boxed{0.9966}$ (after normalization), despite the absence of direct audio–text supervision in this toy.
4. The toy illustrates *possible* transitive alignment, not a mathematical guarantee from pairwise proximity.

Recall: ImageBind → IMAGE-ANCHOR HUB, SIX MODALITIES. Unlike Gemini's mixed-input contextualization, the encoders are aligned in a shared space.

3.10 ColPali (2024)

Original source: ColPali: Efficient Document Retrieval with Vision Language Models ↗

Problem: Turning PDF pages into OCR chunks can discard charts, tables, and layout. Averaging a whole page into one vector can also hide tiny but important regions. **Mechanism:** VLM-powered page-image and query embeddings keep many token/patch vectors (typically projected to 128 dimensions in original ColPali), then use *late interaction* MaxSim per query token.



$$\text{score}(q, D) = \sum_{i=1}^{N_q} \max_{1 \leqslant j \leqslant N_D} \widehat{q}_i^{\top} \widehat{d}_j. \tag{17}$$

Worked example (illustrative, not model measurements)

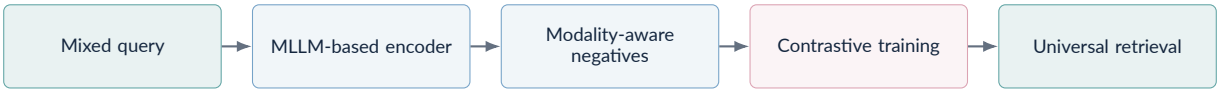
1. Query has two meaningful tokens: “revenue” and “2025”. Page A has patches with similarity rows $[0.9, 0.1, 0.3]$ and $[0.2, 0.8, 0.1]$.
2. MaxSim chooses the best patch for each query token: max $= 0.9$ and max $= 0.8$.
3. score$(q, A) = 0.9 + 0.8 = \boxed{1.7}$. Page B’s best matches are $0.7, 0.2 \Rightarrow$ score$(q, B) = 0.9$.
4. Page A ranks higher even if the relevant table occupies only a small region of its page.

Recall: ColPali → MANY VECTORS PER PAGE, NOT ONE. Cost: larger vector index and more matching operations.

3.11 MM-Embed (2024 / revised 2025)

Original source: MM-Embed: Universal Multimodal Retrieval with Multimodal LLMs ↗

Problem: When adapted naively, multimodal LLM retrievers may favor semantically similar text over genuinely relevant images (*modality bias*). **Mechanism:** contrastively fine-tune an MLLM as a retriever over text/image/mixed tasks, then use modality-aware hard-negative mining and continued training to preserve strong text retrieval.



$$\mathcal{L} = -\log \frac{e^{s(q,d^+)/\tau}}{e^{s(q,d^+)/\tau} + \sum_{d^- \in \mathcal{N}_{\text{mod}}} e^{s(q,d^-)/\tau}}. \tag{18}$$

Worked example (illustrative, not model measurements)

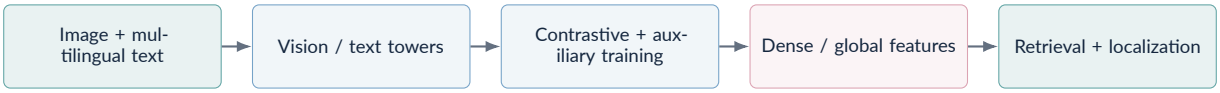
1. Query: “Find the photo that shows this building after renovation.” The correct target is an image; a tempting irrelevant target is a long text article about the same building.
2. Positive score 0.6, misleading hard negative score 0.7, with $\tau = 0.1$: $P_+ = e^6/(e^6 + e^7) = 0.2689$.
3. Loss $= -\ln(0.2689) = \boxed{1.3133}$. This strong penalty motivates fixing the modality bias.
4. If training makes positive score 0.8 and hard-negative 0.5, then $P_+ = 0.9526$, loss 0.0486.

Recall: MM-Embed → FIX MODALITY BIAS USING THE RIGHT HARD NEGATIVES.

3.12 SigLIP 2 (2025)

Original source: SigLIP 2: Multilingual Vision-Language Encoders... ↗

Problem: A global image–text score doesn’t necessarily create precise spatial or multilingual representations. **Training recipe:** extend SigLIP with captioning-oriented pretraining, self-distillation, masked prediction, multilingual data, online curation, and selected flexible-resolution/aspect-ratio configurations.



$$\mathcal{L}_{\text{total}} \approx \lambda_s \mathcal{L}_{\text{sigmoid}} + \lambda_d \mathcal{L}_{\text{distill}} + \lambda_m \mathcal{L}_{\text{masked}} + \lambda_c \mathcal{L}_{\text{caption}}. \tag{19}$$

Worked example (illustrative, not model measurements)

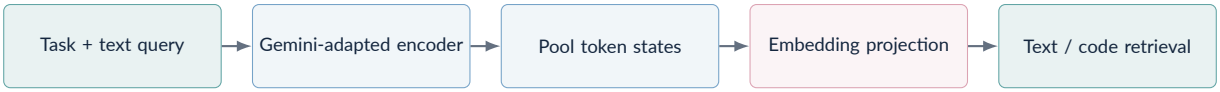
1. A toy training step produces loss components $[0.4, 0.2, 0.3, 0.1]$ for sigmoid, distillation, masked prediction, and caption tasks.
2. If we *illustratively* choose equal weights, total becomes $0.4 + 0.2 + 0.3 + 0.1 = \boxed{1.0}$.
3. Contrastive examples teach cross-modal alignment; masked/dense tasks can encourage more locally useful patch representations; captions introduce linguistic supervision. Real task scheduling and weights must follow the paper.

Recall: SigLIP 2 → MULTILINGUAL + DENSE FEATURES + RICHER TRAINING. The paper describes released vision-tower checkpoints from ViT-B to giant scale; checkpoint parameter counts are not a single fixed number.

3.13 Gemini Embedding (text-only) (2025)

Original source: Gemini Embedding: Generalizable Embeddings from Gemini ↗

Problem: Need versatile multilingual and code-aware text retrieval from pretrained LLM knowledge, not just next-token generation. **Architecture:** an encoder adapted from Gemini, with pooling/projection and multi-stage contrastive training, plus synthetic supervision and task instructions.



$$q = f\big(\mathcal{M}(t \oplus \text{query})\big), \qquad d = f\big(\mathcal{M}(\text{document})\big). \tag{20}$$

Worked example (illustrative, not model measurements)

1. Query "How do I limit API requests?"; positives describe rate limiting; a distractor passage describes database connection pooling.
2. Suppose similarities are 0.75 and 0.25 at $\tau = 0.1$. Then $P_+ = e^{7.5}/(e^{7.5} + e^{2.5}) = 0.9933$.
3. $\mathcal{L} = -\ln(0.9933) = \boxed{0.0067}$.

Recall: 2025 GEMINI EMBEDDING → TEXT / CODE RETRIEVAL FROM A PRETRAINED GEMINI BACKBONE. Not the same thing as the natively multimodal 2026 model.

3.14 Gemini Embedding 2 (2026)

Original source: Gemini Embedding 2: A Native Multimodal Embedding Model from Gemini ↗

Problem: Dual-tower late alignment cannot fully contextualize an instruction with interleaved image, audio and video features before representing the *combined* input. **Architecture:** initialize a multimodal Gemini model, adapt it for bidirectional token-context processing, mean-pool contextualized token states, and apply a newly learned projection to 3,072 dimensions; train additional 768/1,536-dimensional prefixes using MRL.



$$H = \mathcal{M}(T) \in \mathbb{R}^{L \times d_{\mathcal{M}}}, \tag{21}$$

$$\bar{h} = \frac{1}{L}\sum_{i=1}^{L} H_i \in \mathbb{R}^{d_{\mathcal{M}}}, \tag{22}$$

$$E = W\bar{h} + b \in \mathbb{R}^{3072}. \tag{23}$$

Worked example (illustrative, not model measurements)

1. User provides text "find the red bicycle behind the car" plus a photograph. The text and image become interleaved multimodal input tokens, processed in *one* forward pass.
2. For illustration, suppose the transformer returns three 2D contextualized token vectors: $(1,0)$, $(0,1)$, $(2,2)$.
3. Their mean is $\bar{h} = ((1+0+2)/3, (0+1+2)/3) = \boxed{(1,1)}$.
4. For a 2D teaching projection $W = \left[\begin{smallmatrix} 2 & 1 \\ 0 & 1 \end{smallmatrix}\right]$, $b = 0$, the output is $E = (3,1)$. In the real model, the target is 3,072D, not 2D.
5. Candidate photos/videos are embedded *separately*; retrieval then uses cosine similarity between their vectors and this mixed-query vector.

Recall: GEMINI EMBEDDING 2 → FUSE WITHIN THE INPUT, THEN POOL, PROJECT AND RETRIEVE.

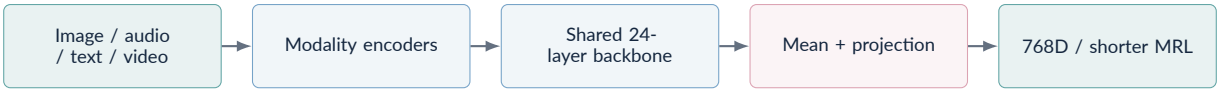
Research caveat

The paper does *not* reveal which exact Gemini backbone or full corpus was used. It confirms reuse of pretrained Gemini, not training the embedding system from scratch; it also does *not* claim that private user chat logs were harvested for training. The architecture above includes only publicly documented steps.

3.15 EmbeddingGemma 2 (2026)

Original source: Official EmbeddingGemma 2 model card ↗

Problem: API-only multimodal retrieval is costly or impossible to run privately on a user's machine. **Architecture:** 740M parameters total (270M text/embedding part, 170M vision encoder, 300M audio encoder); a 24-layer, 512-wide text backbone, mean pooling, then 512-to-768 learned projection. The model supports text, image, video, audio and combinations, plus Matryoshka prefixes at 512/256/128 dimensions.



$$e_{768} = W_{768\times512}\,\bar{h}_{512} + b, \qquad e_{256} = \text{normalize}(e_{768}[1{:}256]). \tag{24}$$

Worked example (illustrative, not model measurements)

1. Index 100,000 local file segments with 768D float32 embeddings. Each vector needs $768 \times 4 = 3{,}072$ raw bytes.
2. Total vector storage $100{,}000 \times 3{,}072 = 307{,}200{,}000$ bytes, or $\boxed{307.2\ \text{MB}}$ (decimal), excluding index and file metadata.
3. Truncate to a trained 256D prefix and re-normalize: storage becomes $100{,}000 \times 256 \times 4 = \boxed{102.4\ \text{MB}}$.
4. That is a 3× reduction; measure any retrieval quality decline on your own dataset.

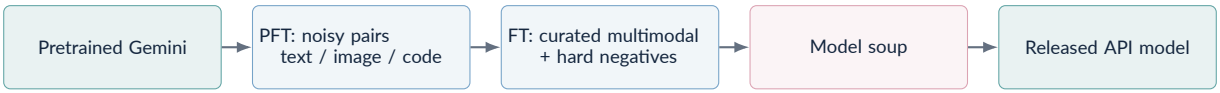
Recall: EMBEDDINGGEMMA 2 → MULTIMODAL EMBEDDINGS YOU CAN RUN LOCALLY. The model card says Apache-2.0 weights, unlike proprietary Gemini Embedding 2.

4. Deep laboratory: rebuilding Gemini Embedding 2 training

4.1 What was actually pretrained and what was learned later?

Component	Evidence-grounded interpretation
Pretrained multimodal knowledge	Inherited from a Gemini checkpoint: the paper explicitly says it initializes from Gemini.
Encoder adaptation	Uses bidirectional attention across the encoded multimodal token sequence.
Pooling	Arithmetic average across contextualized token positions, not summing raw images and words.
Projection	A randomly initialized learned linear map to the final embedding dimension.
Contrastive geometry	Learned using multimodal, single-modality and cross-modal retrieval batches.
Exact backbone details	Undisclosed (parameter count, hidden size, complete training-data recipe).

4.2 Training data and curriculum are separate from user conversations



The pre-fine-tuning stage uses large potentially noisy query–target datasets and relatively large batches. Fine-tuning adds text, code, documents, images, audio and video. Training examples are sampled by task so individual batches generally belong to one task; empirical sampling rates and batch sizes balance the objectives. Model soup averages compatible checkpoints' *parameters*. The paper does not document using consumers' private chats as the training corpus.

4.3 The exact contrastive structure from the paper

For text-only tasks a task instruction t may be concatenated to query text. The operator $\oplus$ denotes sequence concatenation, not numerical addition:

$$\mathbf{q}_i = f\left(\text{meanpool}\left(\mathscr{M}(t \oplus q_i)\right)\right), \quad \mathbf{p}_i^{\pm} = f\left(\text{meanpool}\left(\mathscr{M}(p_i^{\pm})\right)\right). \tag{25}$$

Define $c(x,y) = \cos(x,y)$ and the valid-negative mask

$$m(i,j) = \begin{cases} 0, & q_i = q_j \text{ or } p_i^+ = p_j^+, \\ 1, & \text{otherwise.} \end{cases} \tag{26}$$

Then the paper's general NCE objective can be represented as

$$\mathscr{L} = \frac{1}{B}\sum_{i=1}^{B}\left[-\log \frac{e^{c(\mathbf{q}_i,\mathbf{p}_i^+)/\tau}}{e^{c(\mathbf{q}_i,\mathbf{p}_i^+)/\tau} + e^{c(\mathbf{q}_i,\mathbf{p}_i^-)/\tau} + \sum_{j=1}^{B} m(i,j)e^{c(\mathbf{q}_i,\mathbf{p}_j^+)/\tau}}\right]. \tag{27}$$

The hard-negative term is omitted if no hard negative is supplied. The mask excludes duplicate query/positive entries that should not count as negatives.

Worked example (illustrative, not model measurements)

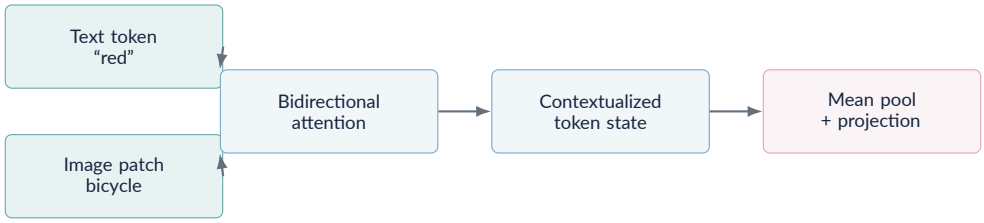
Hand-computed masked contrastive batch. Let $B = 2$. Query q_1 has positive score 0.8, a hard negative 0.5, and one distinct in-batch negative 0.2; take $\tau = 0.1$. For this query:

$$D = e^8 + e^5 + e^2 = 2980.958 + 148.413 + 7.389 = 3136.760,$$
$$P_+ = 2980.958/3136.760 = 0.9503,$$
$$\mathscr{L}_1 = -\log(0.9503) = \boxed{0.0510 \text{ (approximately)}}.$$

If the in-batch target is actually the same positive document (a duplicate), the mask is 0: denominator becomes $e^8 + e^5$, $P_+ = 0.9526$, $\mathscr{L}_1 = 0.0486$. Mistakenly treating a true duplicate as a negative raises the loss for the wrong reason.

4.4 Worked attention example: how image information enters a text token

This tiny example demonstrates the *mechanism* of fusion. It is not a recovered Gemini attention map.



For one text-query token, consider two keys and values, one text and one image:

$$q_{\text{text}} = [1,0], \qquad k_{\text{text}} = [1,0], \qquad k_{\text{image}} = [0,1],$$
$$v_{\text{text}} = [1,0], \qquad v_{\text{image}} = [0,2], \qquad d_k = 2.$$

Compute scaled dot-products:

$$\ell_{\text{text}} = \frac{1}{\sqrt{2}} \approx 0.7071, \qquad \ell_{\text{image}} = \frac{0}{\sqrt{2}} = 0.$$

Apply softmax:

$$\alpha_{\text{text}} = \frac{e^{0.7071}}{e^{0.7071} + e^{0}} \approx 0.6698, \qquad \alpha_{\text{image}} \approx 0.3302.$$

Now mix the *values*:

$$h'_{\text{text}} = 0.6698[1,0] + 0.3302[0,2] = \boxed{[0.6698,\ 0.6604]}. \tag{28}$$

The updated **text-token state now contains a nonzero component from the image value**. Repeating this across heads and layers permits complex cross-modal interactions. In a contrastive embedding model, gradients train the attention and later projection to emphasize relationships that help retrieve true positives over negatives.

4.5 Why multi-dimensional MRL matters

$$\mathscr{L}_{\text{multi}} = \sum_{d \in \{768,1536,3072\}} \lambda_d \mathscr{L}_d(\text{normalize}(E_{1:d})). \tag{29}$$

The equation is an *illustrative combined objective* conveying the overlapping-prefix method; the paper does not disclose all λ_d values. Training every prefix helps one checkpoint serve differently constrained retrieval indexes.

4.6 Model soup is checkpoint arithmetic, not embedding arithmetic

$$\theta_{\text{soup}} = \sum_{k=1}^{K} \alpha_k \theta_k, \qquad \sum_k \alpha_k = 1. \tag{30}$$

Worked example (illustrative, not model measurements)

If the same scalar weight is 2.0 in base checkpoint A and 2.6 in specialized checkpoint B, a 2:1 weighted soup has value $(2\times 2.0 + 1\times 2.6)/3 = \boxed{2.2}$. Every compatible parameter undergoes this operation. No two models are separately queried at inference time.

4.7 What does the evidence actually say?

Benchmark / result	Gemini Embedding 2	Metric
MSCOCO text→image	62.9	Recall@1 (%)
MSCOCO image→text	78.8	Recall@1 (%)
VATEX text→video	68.8	NDCG@10 (%)
ViDoRe V2 document retrieval	64.9	NDCG@10 (%)
MTEB multilingual	69.9	Mean by task
MTEB code	84.0	Mean by task

Scores above come from the original paper and are *not directly interchangeable* across metrics or datasets.

On ViDoRe V2, one competing specialized Voyage model reports 65.5, above Gemini's 64.9. The paper establishes a strong full system, but does not isolate exact quantitative gains from fusion alone.

Research caveat

The code retrieval ablation reports means of 70.5 (older Gemini embedding), 73.0 (Embedding 2 without synthetic data), and 86.3 (with synthetic data) on three selected tasks. Thus adding synthetic data within the new model improves by **13.3** points; relative to the older model, the gain is 15.8 points. These are distinct comparisons.

5. Retrieval evaluation: every metric, fully calculated

This chapter repairs the typesetting issue in the previous edition: all formulas here are displayed using mathematical typesetting, and every metric has a hand-computed example.

5.1 Vocabulary before scoring anything

A **query** is an input asking for relevant results. A **candidate** is an indexed text, image, clip, or page. **Relevance labels** come from a benchmark's ground truth, not from cosine similarity. Rank 1 is the top returned result. The cutoff K is the number of inspected results.

> **Cosine similarity** scores *one query–candidate pair*. **Recall@K**, **MRR@K**, and **NDCG@K** evaluate *ranked retrieval results against known relevance labels*. Never infer retrieval accuracy from a cosine value alone.

5.2 Recall@K: how many relevant results did the top K retrieve?

For a single query with relevant set R and top-K result set S_K,

$$\text{Recall}@K = \frac{|R \cap S_K|}{|R|}. \tag{31}$$

For single-positive benchmarks a common aggregated form is

$$R@K = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}[r_i \leqslant K], \tag{32}$$

where r_i is the rank of the query's only correct target. The first equation is *per-query set recall*; the second is a *query-level success fraction*, which coincides with recall when each query has exactly one relevant target.

Worked example (illustrative, not model measurements)

Example A – multiple relevant items. A query has four genuinely relevant images: $R = \{I_2, I_4, I_8, I_9\}$. The system returns top 5: $S_5 = [I_8, I_3, I_2, I_7, I_1]$.

1. Intersection $R \cap S_5 = \{I_8, I_2\}$ (two relevant images found).
2. Total relevant images in the collection: $|R| = 4$.
3. Recall@5 = 2/4 = $\boxed{0.50 = 50\%}$.

Example B – single target per query. Among 100 independent text queries, 63 retrieve the correct image at rank 1. Then $R@1 = 63/100 = \boxed{63\%}$, *not* a cosine of 0.63.

5.3 Precision@K and Hit@K: separate but useful

$$\text{Precision}@K = \frac{|R \cap S_K|}{K}, \qquad \text{Hit}@K = \mathbb{1}\big[|R \cap S_K| > 0\big]. \tag{33}$$

In Example A above, Precision@5 = 2/5 = $\boxed{40\%}$, Hit@5 = 1. Recall@5 is 50%. Different questions, different denominators.

5.4 MRR@K: how high is the first correct result?

For query i, define r_i as the rank of the *first* relevant candidate. Its reciprocal rank at cutoff K is zero if no correct candidate appears within the cutoff:

$$\boxed{\text{MRR}@K = \frac{1}{N}\sum_{i=1}^{N} \begin{cases} \dfrac{1}{r_i}, & r_i \leqslant K, \\ 0, & r_i > K. \end{cases}} \tag{34}$$

Worked example (illustrative, not model measurements)

Three-query worked example at $K = 10$. The first relevant results are at ranks $[1, 2, 5]$.

1. Query 1: reciprocal rank $1/1 = 1.0000$.
2. Query 2: reciprocal rank $1/2 = 0.5000$.
3. Query 3: reciprocal rank $1/5 = 0.2000$.
4. Sum $= 1.7$. Average over *three queries*: $1.7/3 = \boxed{0.5667}$.

$$\text{MRR@10} = \frac{1}{3}\left(1 + \frac{1}{2} + \frac{1}{5}\right) = \boxed{0.5667}.$$

Counterexample: If query 3's first correct hit is at rank 15, it contributes zero at @10. MRR becomes $(1 + 0.5 + 0)/3 = \boxed{0.5}$. MRR ignores any later relevant items after the first hit.

5.5 DCG@K: graded relevance, with diminishing value by rank

When multiple items can be differently relevant, graded relevance is appropriate. A common exponential-gain convention is:

$$\boxed{\text{DCG@}K = \sum_{i=1}^{K} \frac{2^{\text{rel}_i} - 1}{\log_2(i+1)}}. \tag{35}$$

Different evaluation frameworks may use different gain conventions; match the benchmark before comparing numbers.

Worked example (illustrative, not model measurements)

Suppose the top three items have relevance labels $[3, 0, 2]$.

Rank i	rel_i	Gain $2^{\text{rel}_i} - 1$	Discount $\log_2(i+1)$	Contribution
1	3	7	1	$7/1 = 7$
2	0	0	$\log_2 3 \approx 1.585$	0
3	2	3	2	$3/2 = 1.5$

Therefore DCG@3 $= 7 + 0 + 1.5 = \boxed{8.5}$.

5.6 NDCG@K: DCG compared with the best possible ordering

$$\boxed{\text{NDCG@}K = \frac{\text{DCG@}K}{\text{IDCG@}K}}, \tag{36}$$

where IDCG is DCG after sorting relevant items in ideal descending relevance order. If the ideal DCG is zero, the benchmark must define how that query is handled (often omitted or scored zero).

Worked example (illustrative, not model measurements)

Continue the actual ranking $[3, 0, 2]$, which had DCG@3 $= 8.5$.

1. Reorder by relevance to get the ideal ranking $[3, 2, 0]$.
2. Ideal contributions: rank 1 $= (2^3 - 1)/\log_2 2 = 7$; rank 2 $= (2^2 - 1)/\log_2 3 = 3/1.58496 \approx 1.89279$; rank 3 $= 0$.
3. IDCG@3 $= 7 + 1.89279 = \boxed{8.89279}$.
4. Normalize: NDCG@3 $= 8.5/8.89279 = \boxed{0.95583}$.

5.7 Text-to-image vs image-to-text: what actually changes?

A direction is not a reciprocal probability. The *query set*, *candidate collection*, and often the count of positives per query differ.



Worked example (illustrative, not model measurements)

Miniature candidate matrix (illustrative cosine scores):

$$S_{\text{images}\times\text{captions}} = \begin{bmatrix} 0.91 & 0.22 & 0.12 \\ 0.16 & 0.74 & 0.80 \end{bmatrix}.$$

Let row 1 be image A, row 2 image B; columns are captions C1 (A), C2 (B), and C3 (irrelevant). Image→text: A retrieves C1 correctly; B retrieves C3 incorrectly, so $R@1 = 1/2 = 50\%$. Text→image: C1 retrieves A correctly; C2 retrieves B correctly. Evaluated on the two target-caption queries, $R@1 = 2/2 = 100\%$. **Same matrix, opposite direction, different scores.**

5.8 Which metric belongs to which benchmark?

Typical task	Frequent metric	Reason
Text–image / image–text	Recall@1, Recall@5	Was a valid paired target near the top?
Visual documents	NDCG@5 or @10	Graded relevance; several pages may be useful.
Audio-text / speech retrieval	MRR@10	Reward finding the first good passage early.
Video retrieval	NDCG@10	Reward relative relevance throughout the ranked list.

Research caveat

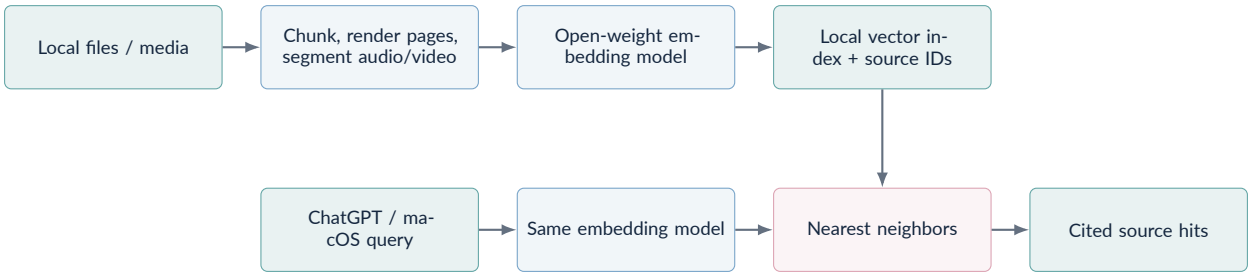
The particular MSCOCO scores 62.9 (text→image) and 78.8 (image→text) in Gemini Embedding 2 are from the paper's published evaluation protocol. The toy similarity matrix above is *not* an explanation of those exact values or a reproduction of the original experiment.

6. From training to a working retrieval product

6.1 Training versus inference: don't confuse the operations

	Training time	**Query time**
Inputs	Many labeled or weakly labeled query / positive / negative tuples	A single user query + precomputed corpus vectors
Main computation	Backpropagate contrastive loss into encoder weights	Forward-pass query embedding, nearest-neighbor search
Output	Updated encoder parameters θ	Top-K IDs, similarity scores, source metadata
What is fixed?	Dataset labels/positive relation define supervision	Trained encoder and indexed vector space

6.2 Architecture for a local macOS app and optional MCP plugin



Recommended first benchmark: (1) 10–20 mixed-content PDFs, screenshots and audio snippets; (2) 30 queries with manually recorded relevant source IDs; (3) compare one-vector multimodal retrieval with text-only OCR/transcription; (4) record Recall@5, MRR@10, NDCG@10, latency and memory; (5) add optional reranking only if retrieval quality requires it.

6.3 Memory estimate for raw float32 vectors

$$\text{bytes} = N \times d \times 4. \tag{37}$$

Dimensions	Bytes/vector	Raw storage for 10^6 vectors
128	512	0.512 GB
256	1,024	1.024 GB
768	3,072	3.072 GB
1,536	6,144	6.144 GB
3,072	12,288	12.288 GB

These numbers exclude ANN index overhead, original files, metadata, caching, and potential compression. Embeddings from two different model spaces cannot be assumed compatible even if both are 768D.

6.4 Open versus proprietary: licensing is about artifacts

Artifact	**Practical licensing position**
Gemini Embedding 2 research paper	ArXiv indicates CC BY 4.0 for its text/figures with attribution and license compliance.
Gemini Embedding 2 trained weights	No downloadable public open-weight license verified; API terms and pricing govern usage.
EmbeddingGemma 2 model weights	Official Google model card identifies Apache 2.0. Check the exact repository/license when distributing.
This educational handbook	Independent explanations, formulas, numerical teaching examples and vector reconstructions; cites original research.

7. Glossary and active recall

Dual encoder: two separate towers whose output vectors are aligned.
Fusion: interaction among modality tokens before encoding is summarized.
Contrastive learning: optimize matching examples to score above negatives.
Hard negative: misleadingly close but incorrect item.
Pooling: reduce variable-length token representations to fixed-size output.
Projection: learned transformation into output vector coordinates.
MRL: train shorter prefixes of a long embedding to be useful.
Late interaction: compare retained token/patch vectors at ranking time.
Recall@1: fraction of queries with a relevant result at rank one (single-positive setup).
MRR: mean reciprocal rank of each query's first relevant hit.
NDCG: normalized graded relevance after logarithmic rank discount.
Model soup: parameter-average compatible fine-tuned checkpoints.

Core intuition

Ten retrieval and architecture prompts: (1) Derive CLIP's two directional losses. (2) Why does ALIGN need noisy scale? (3) Why is SigLIP a loss innovation? (4) What does CoCa's captioning branch learn? (5) How does ImageBind avoid every cross-pair dataset? (6) Compute ColPali MaxSim for two tokens. (7) Explain Gemini's interleaved token sequence, bidirectional attention and pooling. (8) Describe the three training stages and the role of synthetic data. (9) Calculate Recall@5, MRR@10 and NDCG@3 from relevance ranks. (10) Design a benchmark to compare EmbeddingGemma 2 locally to a text-only retrieval baseline.

8. Parameter count is not embedding dimension

Model / config	Illustrative output width	Interpretation / limitation
CLIP ViT-B/32	512D	Checkpoint-specific; not every CLIP is 512D.
ColPali original	128D per token/patch vector	Many vectors per page, not one 128D vector per PDF.
Gemini Embedding 2	3072D; MRL 1536D, 768D	Backbone parameter count undisclosed.
EmbeddingGemma 2	768D; MRL 512D, 256D, 128D	740M trained parameters total; different from output width.
SigLIP / SigLIP 2	Checkpoint-dependent	Vision-tower parameter size is not embedding vector width.

9. Original publications and primary documentation

All links below are clickable in the PDF. Sources are included for verification, not to imply every trained checkpoint has the same license as its paper. The history and reported experimental numbers are derived from the authors' original materials.

Paper / model	Year	Primary source
SimCLR	2020	https://arxiv.org/abs/2002.05709
CLIP	2021	https://arxiv.org/abs/2103.00020
ALIGN	2021	https://arxiv.org/abs/2102.05918
AudioCLIP	2021	https://arxiv.org/abs/2106.13043
VideoCLIP	2021	https://arxiv.org/abs/2109.14084
CLAP	2022	https://arxiv.org/abs/2206.04769
CoCa	2022	https://arxiv.org/abs/2205.01917
SigLIP	2023	https://arxiv.org/abs/2303.15343
ImageBind	2023	https://arxiv.org/abs/2305.05665
ColPali	2024	https://arxiv.org/abs/2407.01449
MM-Embed	2024	https://arxiv.org/abs/2411.02571
SigLIP 2	2025	https://arxiv.org/abs/2502.14786
Gemini Embedding	2025	https://arxiv.org/abs/2503.07891
Gemini Embedding 2	2026	https://arxiv.org/abs/2605.27295
EmbeddingGemma 2	2026	https://ai.google.dev/gemma/docs/embeddinggemma/model_card_2
EmbeddingGemma 2 weights	2026	https://huggingface.co/google/embeddinggemma-2
Gemini API embedding guide	2026	https://ai.google.dev/gemini-api/docs/embeddings
Gecko: synthetic-data text embedding	2024	https://arxiv.org/abs/2403.20327
NV-Embed: LLMs as embedders	2024	https://arxiv.org/abs/2405.17428
Matryoshka Representation Learning	2022	https://arxiv.org/abs/2205.13147
Contrastive Predictive Coding (NCE)	2018	https://arxiv.org/abs/1807.03748

LAST-PAGE MEMORY TRACE

SIMILARITY → ALIGN (CLIP) → SCALE (ALIGN) → ADD SOUND / TIME (AudioCLIP, VideoCLIP, CLAP) → CAPTION TOO (CoCa) → OPTIMIZE PAIRS (SigLIP) → ALIGN SIX MODALITIES (ImageBind) → KEEP PATCHES (ColPali) → FIX BIAS (MM-Embed) → IMPROVE MULTILINGUAL DENSE FEATURES (SigLIP 2) → FUSE MIXED TOKENS (Gemini Embedding 2) → RUN LOCALLY (EmbeddingGemma 2).